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CGI - A Comprehensive Framework for Classifying Biotic and Abiotic Stress in Plants

Tesi di Laurea Magistrale in
Telecommunication Engineering

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Abstract

Plant stress identification is a cornerstone of precision agriculture. This project develops a robust binary classification pipeline to distinguish between **Biotic** and **Abiotic** stresses across 8 distinct crop species. To overcome the "resolution bottleneck" caused by standard image resizing, we implemented an **Adaptive Tile-based Framework** combined with a **Soft Voting** ensemble logic. This approach preserves microscopic pathological textures by processing high-resolution image patches independently.

Experimental results indicate that the optimal model (at **Epoch 12**) achieves a weighted **F1-score of 0.95** and a peak leaf-level accuracy of **95.5%** on the internal test set. In cross-species inference tasks, the model maintains a robust **83.97%** accuracy, demonstrating significant generalization potential across unseen cultivars. This report details the customized **MobileNetV2** architecture, the implementation of **Automatic Class Weighting** for data imbalance, and provides visual evidence of diagnostic reliability through **Grad-CAM** attention analysis.

Keywords: Plant Stress, MobileNetV2, Tile-based Voting, Soft Voting, Grad-CAM, PyTorch.

1 | Introduction

1.1. Background and Motivation

Agricultural productivity is increasingly threatened by various stressors that contribute to substantial global economic losses [1]. These are fundamentally classified into **Biotic stress** (e.g., fungal, bacterial, and pest infestations) and **Abiotic stress** (e.g., nutritional deficiencies and environmental imbalances). In precision agriculture, the ability to rapidly and accurately differentiate these two categories is critical; it dictates the immediate intervention strategy, such as applying specific fungicides versus adjusting fertilization regimes. An incorrect diagnosis not only leads to catastrophic crop failure but also results in the misuse of chemical treatments, further degrading soil health and increasing environmental toxicity.

1.2. Problem Statement

While Deep Learning (DL) has shown immense promise in automated plant phenotyping, its practical deployment in field conditions is constrained by several technical bottlenecks. This research investigates the following key challenges:

- **Visual Mimicry and Diagnostic Ambiguity**: Differentiating symptoms that exhibit high visual similarity, such as leaf chlorosis (yellowing) caused by both nitrogen deficiency (Abiotic) and early-stage fungal infection (Biotic).
- **Resolution Bottleneck**: Standard preprocessing methods that involve global image resizing often lead to the loss of microscopic textural details, rendering early-stage lesions statistically indistinguishable from background noise.
- **Data Imbalance**: Managing the inherent scarcity of expertly labeled abiotic stress samples compared to the relatively abundant biotic datasets, which often biases model predictions.
- **Domain Shift and Generalization**: Ensuring that a model trained on a specific

subset of cultivars remains robust and reliable when deployed on unseen species or in varying environmental lighting.

1.3. Project Objectives

Building on the principles of transfer learning and mobile-efficient architectures [2], this project develops a robust binary diagnostic system. To overcome the "Resolution Bottleneck," we propose an **Adaptive Tile-based Voting Framework** that focuses on localized pathological textures rather than global leaf geometry.

The core objectives are:

1. To implement an **Adaptive Preprocessing Pipeline** that preserves localized lesion features through 336-pixel short-side resizing and 50% overlapping tiling.
2. To develop a **Soft-voting Ensemble Logic** that aggregates tile-level probabilities into a final, high-confidence leaf-level diagnosis.
3. To mitigate class imbalance through **Automatic Weight Compensation** in the loss function, ensuring balanced sensitivity for both stress types.
4. To evaluate the **Cross-species Robustness** of the model, specifically assessing its performance on unseen data to simulate real-world field conditions.

2 | Literature Review

2.1. Evolution of Plant Stress Diagnosis

The transition from manual scouting to automated diagnostic systems has been accelerated by advancements in optical sensing. As reviewed by Gao et al. [1], RGB and hyperspectral imaging are now integrated with Deep Learning (DL) to identify physiological changes. Unlike traditional machine learning which relies on hand-crafted features, Convolutional Neural Networks (CNNs) automatically learn hierarchical representations of stress symptoms, significantly improving diagnostic accuracy in complex field environments.

2.2. Deep Learning Architectures in Imaging Analysis

According to the foundational logic presented by Gao et al. [1], two typical neural network architectures are prevalent in agricultural imaging analysis, as illustrated in Figure 2.1:

- **Autoencoders (a)**: Primarily used for unsupervised data reconstruction or dimensionality reduction where the input and output dimensions remain identical.
- **Convolutional Neural Networks (b)**: This architecture forms the **methodological core of our project**. It utilizes a supervised learning flow where input pixels pass through deep layers for spatial feature extraction. In **binary classification** tasks, the final dense layer is typically configured with two nodes to represent the probability distribution between distinct classes (e.g., Biotic vs. Abiotic).

2.3. Transfer Learning and Model Efficiency

The efficiency of DCNNs is often enhanced through transfer learning. Leveraging pre-trained architectures such as **MobileNetV2** allows models to achieve high performance even on limited agricultural datasets by reusing feature extractors learned from massive

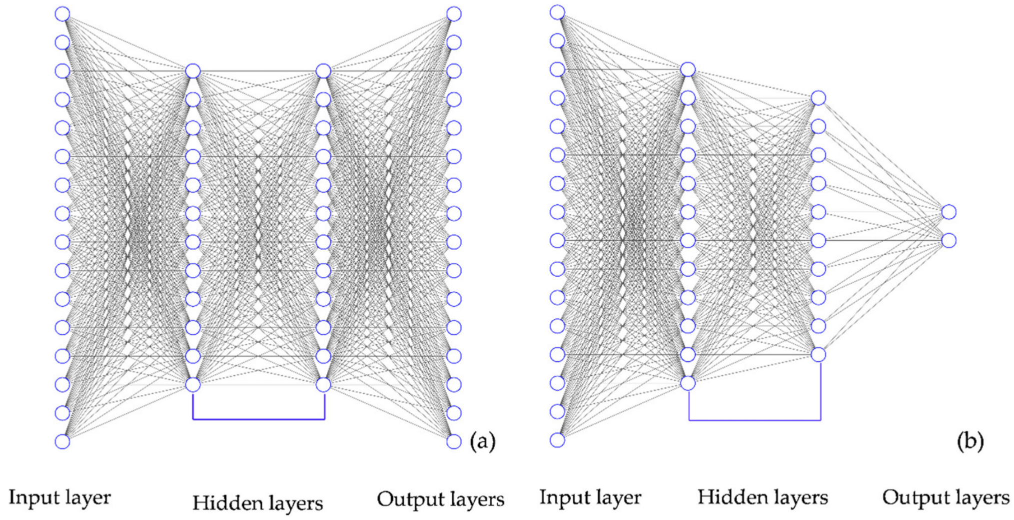


Figure 2.1: Typical architectures of deep neural networks: (a) Autoencoder for unsupervised reconstruction; (b) Convolutional Neural Network for supervised classification [1].

datasets like ImageNet [2]. This paradigm serves as the backbone of our study, enabling the capture of complex pathological features with significantly reduced computational overhead, making it suitable for potential mobile deployment.

2.4. Tile-based Strategy vs. Global Classification

A critical limitation in standard CNNs is the "Resolution Bottleneck"—the loss of fine-grained details during global image resizing. Recent literature suggests that **Tile-based (or Patch-based) strategies** mitigate this by preserving the native resolution of local segments. By analyzing multiple localized tiles and employing an **ensemble voting mechanism**, models can focus on localized "regions of interest" (ROIs)—such as microscopic fungal lesions—more effectively than global models which often suffer from feature blurring.

2.5. Challenges: Domain Adaptation and Class Imbalance

Despite advancements, two major gaps persist in current agricultural AI research:

1. **Domain Adaptation**: Model performance often degrades when applied to species or environments not present in the training set (domain shift). This necessitates testing on unseen data to evaluate universal applicability.

2. **Data Imbalance**: Abiotic stress samples are frequently underrepresented compared to biotic images. Literature emphasizes the necessity of algorithmic compensations, such as **Weighted Loss Functions** or **Automatic Class Weighting**, to prevent the model from developing a bias toward the majority class.

2.6. Summary: Bridging the Gap

While existing studies provide a strong foundation for plant disease detection, there is a lack of integrated systems that combine high-resolution tile-based analysis with automated class-imbalance compensation across multiple crop species. Our study addresses this by developing a MobileNetV2-based framework that utilizes **Soft Voting** and **Class Weighting** to ensure robust binary diagnosis between Biotic and Abiotic stresses.

3 | Methodology

3.1. Problem Definition and Dataset

The study utilizes a large-scale multi-species dataset comprising approximately 20,000 RGB images across 8 distinct crops: Chilli, Tomato, Carambola, Guava, Cherry, Jackfruit, Maize, and Wheat.

3.1.1. Binary Stress Taxonomy

The primary objective is binary classification: **Biotic Stress** (pathogen-induced) and **Abiotic Stress** (environmental). To maintain a focused diagnostic boundary, **healthy leaf samples were excluded from the training phase**, leaving only symptomatic data. This configuration forces the model to optimize its decision boundary purely on the distinctive textures of stress manifestations, preventing the "absence of symptoms" from being exploited as a classification shortcut.

- **Biotic Stress**: Encompasses damage from living organisms, including pathogens (fungal spots, bacterial blights, viral infections) and pests (leaf miners, chewing insects).
- **Abiotic Stress**: Covers non-living environmental factors, primarily nutritional deficiencies (N, P, K, Mg, or Fe deficiency) and physiological senescence.

3.1.2. Visual Representation of Stresses

The diagnostic challenge stems from the high degree of visual mimicry between stressors. As illustrated in Figure 3.1, while a healthy leaf serves as the morphological baseline, Biotic and Abiotic paths lead to distinct yet overlapping textural manifestations.

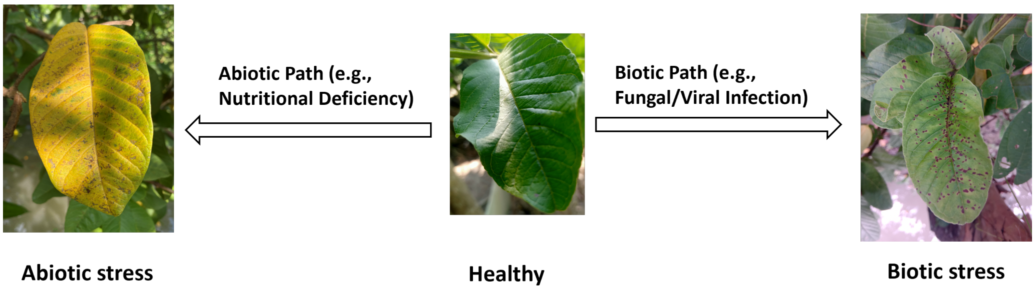


Figure 3.1: Visual manifestation of plant stress categories: (Left) Abiotic stress; (Right) Biotic stress; (Center) Healthy baseline.

3.2. Proposed Model Architecture: Custom MobileNetV2

The diagnostic engine is a modified **MobileNetV2** architecture, selected for its high efficiency and suitability for mobile agricultural deployment. The model follows a 10-block feature extraction hierarchy (Figure 3.2).

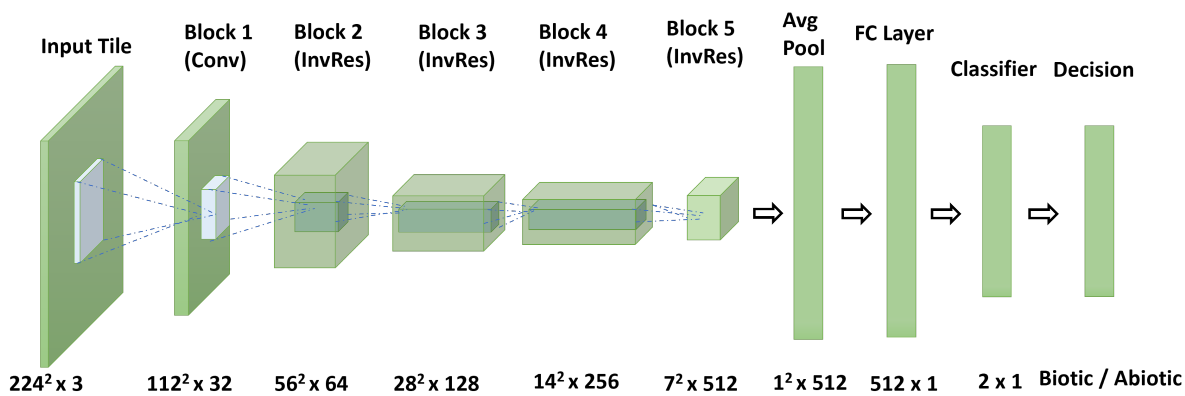


Figure 3.2: Proposed diagnostic pipeline: From adaptive tiling, through a 10-block feature extraction hierarchy, to final soft voting.

The core of the architecture is the **Inverted Residual Block**, which utilizes a "narrow-wide-narrow" pipeline to capture intricate pathological textures through **Pointwise Expansion**, **Depthwise Convolution**, and **Linear Bottlenecks**. This structure ensures that high-dimensional spatial features are preserved while maintaining computational efficiency.

3.2.1. Hierarchical Feature Extraction (Blocks 1–5)

The model processes individual 224×224 tiles through five critical stages:

- **Input Stage (Block 1)**: Initial feature sampling from raw RGB tiles.

- **Spatial Reduction (Blocks 2–3)**: Standard convolutions identify low-level primitives such as edge gradients and localized color intensities.
- **Deep Integration (Blocks 4–5)**: Kernels transition to **embedded volumetric integration**, synthesizing high-dimensional patterns across 512 channels. The spatial resolution is progressively reduced to 7×7 to concentrate information density.

3.2.2. Global Synthesis and Binary Output (Blocks 7–10)

- **Pathological Fingerprinting**: Blocks 7 and 8 utilize Global Average Pooling (GAP) to collapse spatial dimensions into a 512-dimensional vector, representing a condensed "pathological fingerprint" of the tile.
- **Binary Classification**: Block 9 is re-engineered with a **2-node output layer**, mapping the extracted features specifically onto the probability distribution of Biotic and Abiotic categories.

3.3. Experimental Implementation Details

3.3.1. Image Preprocessing Evolution: Global Padding vs. Overlapping Tiling

To evaluate the impact of spatial resolution on diagnostic accuracy, two distinct preprocessing pipelines were developed:

1. **Global Padding Strategy (Baseline)**: Input images were resized by scaling the long side to 256 pixels, with the short side padded with black pixels to maintain a square aspect ratio. This method preserves global leaf geometry but results in significant compression of localized pathological textures.
2. **Adaptive Overlapping Tiling (Proposed)**: The short side of the image was resized to 336 pixels. The image was then partitioned into 224×224 tiles using a **50% overlap ratio**. This strategy ensures that high-resolution textural details are preserved and that lesions located at tile boundaries are captured across multiple spatial views.

3.3.2. Adaptive Tile-based Partitioning

To maintain computational efficiency while preserving resolution, each image generates **up to $N = 8$ tiles**. This mechanism filters out irrelevant background noise and forces

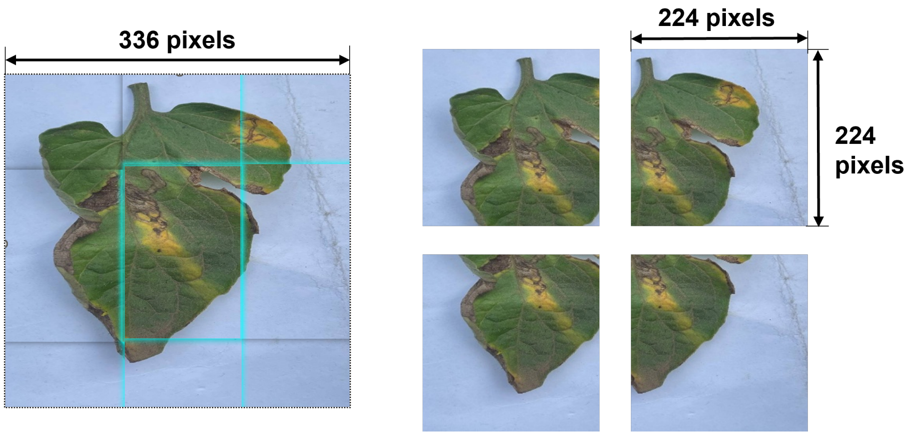


Figure 3.3: Adaptive Tiling Strategy

the model to focus on localized lesion details.

3.3.3. Soft Voting Decision Logic

To convert the raw logits from the final fully connected layer (Block 9) into a valid probability distribution, the **Softmax function** is applied. For a given tile x, the probability of it belonging to class i is defined as:

$$P(y = i|x; \Theta) = \mathrm{softmax}(x; \Theta) = \frac{e^{(W_i^L)^T x + b_i^L}}{\sum_{k=1}^{K} e^{(W_k^L)^T x + b_k^L}} \tag{3.1}$$

where W^L and b^L represent the weights and biases of the output layer, and $K = 2$ for our binary classification task.

Leaf-Level Ensemble Inference

The final diagnostic result for a single leaf is determined by aggregating the confidence scores of all generated tiles. As illustrated in our inference process (see Table 3.1), the model calculates the probability for each individual tile.

For instance, even if specific tiles exhibit lower confidence or localized ambiguity, the **arithmetic mean** of the probability vectors provides a noise-resistant decision. Following the calculation logic shown in our experimental logs:

- **Tile-level probabilities**: $[P_1, P_2, P_3, P_4] = [0.9, 0.8, 0.1, 0.5]$ (Biotic scores)
- **Ensemble Calculation**: $L_{Biotic} = \frac{0.9+0.8+0.1+0.5}{4} = 0.575$

Table 3.1: Example of Soft Voting Ensemble: Probabilities determined for four tiles and aggregated for the final leaf-level classification.

Tile No.	P_{Biotic}	$P_{Abiotic}$
Tile 1	0.90	0.10
Tile 2	0.80	0.20
Tile 3	0.10	0.90
Tile 4	0.50	0.50
Average Probability	**0.575**	**0.425**
Final Decision	**Biotic**	

The final result is thus categorized as **Biotic** since $L_{Biotic} > 0.5$. This soft-voting approach leverages the "Confidence Level" of each tile, ensuring that the model's decision is based on a global synthesis of local pathological features.

Once these individual tile probabilities are obtained, the final plant-level diagnosis aggregates the results of N constituent tiles. The ensemble probability L_j for class j is calculated as the mean confidence score:

$$L_j = \frac{1}{N} \sum_{i=1}^{N} P_{i,j} \tag{3.2}$$

The final classification is determined by selecting the class with the highest ensemble probability: Result $= \arg\max_{j \in \{0,1\}}(L_j)$.

3.3.4. Class Weight Compensation

To mitigate the 10:1 data imbalance between categories, **Automatic Class Weights** (W_c) are applied to the Cross-Entropy Loss function:

$$W_c = \frac{N_{total}}{2 \times N_c} \tag{3.3}$$

This weights the loss of the minority class (Abiotic) more heavily, optimizing for a balanced sensitivity across the diagnostic system.

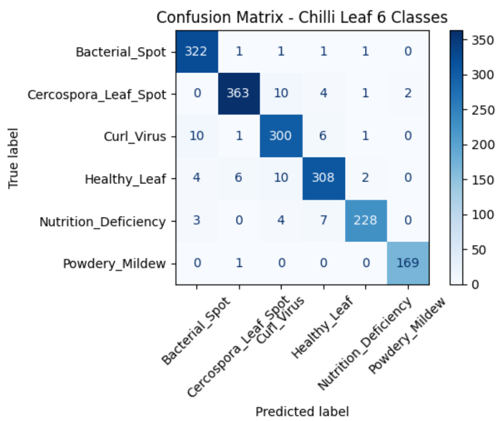
4 | Experiments and Results

4.1. Experimental Setup

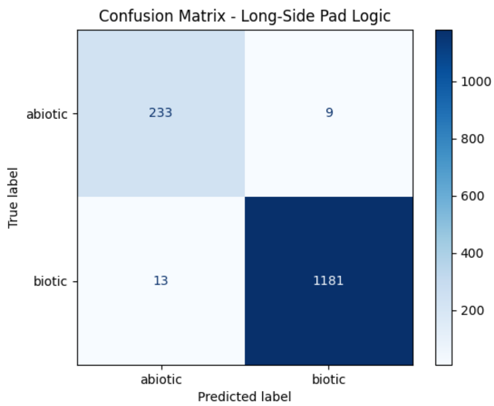
The diagnostic system was implemented using the **PyTorch** framework. The core architecture, *get_plant_model*, was trained using an Adam optimizer with an initial learning rate of 10^{-4}. Based on the comparative analysis of validation loss and leaf-voted stability, the weights from **Epoch 12** were identified as the optimal state for final deployment.

4.2. Phase 1: Preliminary Case Study on Chilli

The project originated from a focused study on the Chilli dataset to verify the model's ability to differentiate between specific stress symptoms before scaling to multiple species.



Chilli: 6-class Specific Classification



Chilli: Binary Stress Classification

Figure 4.1: Experimental results on the Chilli dataset: (Left) Detailed diagnosis of 6 disease/stress categories; (Right) Simplified binary classification for Biotic vs. Abiotic stress.

Analysis: As shown in Figure 4.1, the model achieved high precision in granular classification. By aggregating these 6 specific categories into a binary framework, the system demonstrated near-perfect diagnostic reliability for the Chilli species, providing a solid foundation for cross-species expansion.

4.3. Phase 2: Optimized Tile-based Strategy (Multi-Species Expansion)

Following the successful pilot study on Chilli, the diagnostic scope was expanded to a comprehensive multi-species dataset to evaluate the model's universal applicability. This expanded dataset encompasses "8 distinct crop species", including Chilli, Tomato, Carambola, Guava, Cherry, Jackfruit, Maize, and Wheat. As illustrated in Figure 4.2, these species present a wide array of morphological variations and diverse pathological manifestations, ranging from broad-leaf dicots to narrow-leaf monocots.

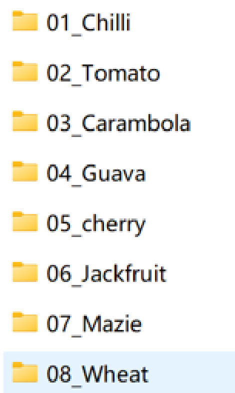


Figure 4.2: Representative samples from the 8-species dataset.

To maintain high-resolution diagnostic accuracy across these diverse species, the **Adaptive Tile-based Strategy** was implemented. The final pipeline employed **336-pixel short-side resizing** and **50% overlapping tiles** (224×224). This method, combined with a **Soft Voting** mechanism, ensures that fine-grained pathological textures—rather than species-specific leaf geometry—are preserved and evaluated collectively for the final leaf-level diagnosis.

4.3.1. Performance on Internal Test Set

The model demonstrated exceptional performance on the internal test set. As illustrated in Figure 4.3, the **leaf-level accuracy** (calculated via Soft Voting) reached its peak and stabilized at **Epoch 12**, achieving an overall accuracy of **95.5%**.

The confusion matrix for this optimal state is shown in Figure 4.4. Despite the visual mimicry between certain Abiotic deficiencies and Biotic symptoms, the model maintains

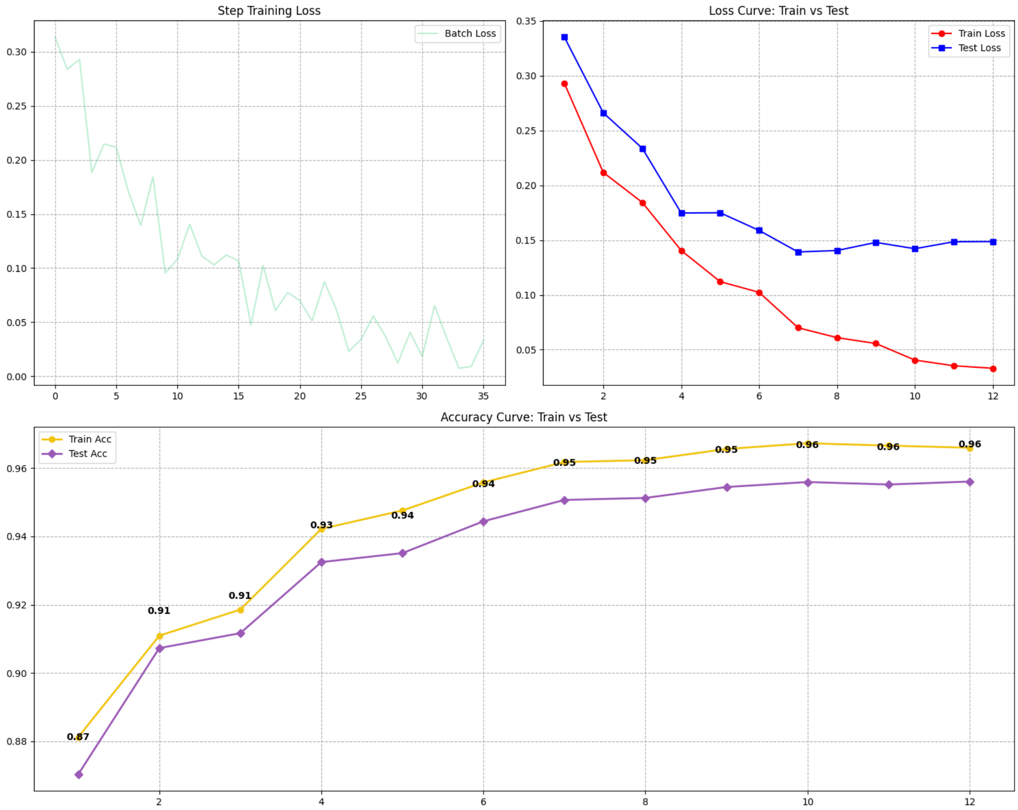


Figure 4.3: Accuracy and Loss curves for the Tile-based Diagnostic Method. The leaf-voted accuracy (blue line) demonstrates superior stability, reaching its optimal state at Epoch 12.

high diagnostic sensitivity with minimal misclassification.

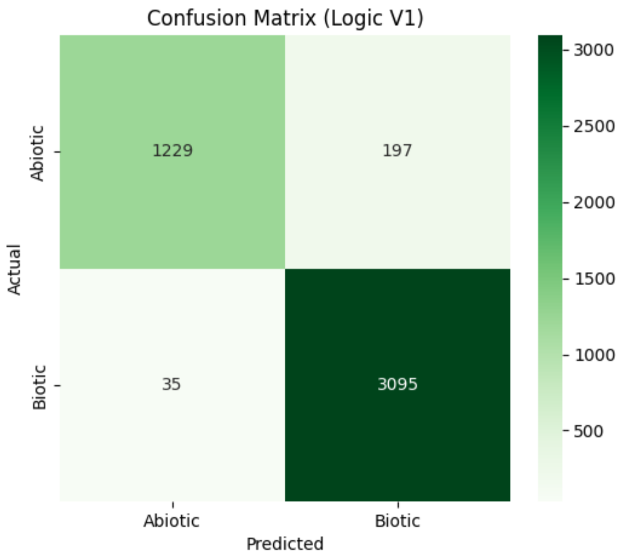


Figure 4.4: Final Confusion Matrix for the Tile-based Diagnostic System (Epoch 12) on Internal Test Set (95.5% Accuracy).

4.3.2. Cross-Species Robustness and Threshold Analysis

To evaluate the model's potential for real-world deployment, testing was conducted on an **unseen species dataset** (Cross-Species Inference). This task yielded a robust accuracy of **83.97%** at Epoch 12. Table 4.1 demonstrates the model's performance across various decision thresholds (τ).

Note that as the threshold τ increases to 0.95 (Strict), the number of correctly identified Abiotic cases rises to **149**, which is significant for reducing false-positive biotic alarms in specific agricultural scenarios.

Table 4.1: Inference Performance Comparison across Thresholds (Epoch 12) on Unseen Species.

Threshold τ	Overall Accuracy	Abiotic Correct	Biotic Correct	F1-Score
0.5 (Standard)	**83.97%**	99	**4118**	**0.8433**
0.6	81.82%	102	4007	0.8299
0.7	79.51%	111	3882	0.8161
0.8	76.22%	117	3711	0.7953
0.95 (Strict)	70.05%	**149**	3369	0.7557

4.3.3. Visual Explainability (Grad-CAM)

To validate that the model's predictions are based on biologically relevant symptoms rather than background noise, the **Grad-CAM (Gradient-weighted Class Activation Mapping)** technique was utilized. As illustrated in Figure 4.5, the visual attention of the model evolves through the network's layers at Epoch 12:

- **High-Attention Regions (Red/Orange)**: These regions represent the highest pixel contribution to the final classification decision. In Layer 16, these hot spots precisely overlap with necrotic lesions, fungal spots, and chlorotic tissues, confirming that the model identifies specific "pathological fingerprints."
- **Low-Attention Regions (Blue/Purple)**: These areas indicate regions with minimal to zero influence on the decision-making process. The model effectively treats healthy green tissue and soil background as neutral noise, assigning them low activation values.

As the feature extraction progresses from the initial layers (Layer 0) to the deeper bottlenecks (Layer 16), the focus transitions from the global leaf contour to localized diagnostic textures. This alignment between the model's heatmaps and actual biological stress manifestations ensures the transparency and reliability of the diagnostic system for real-world agricultural use.

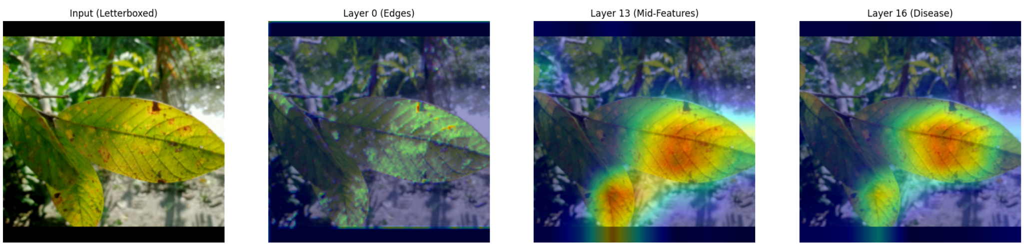


Figure 4.5: Grad-CAM visualization for Epoch 12: Heatmap transitions from global contours (Layer 0) to precise lesion localization (Layer 16).

5 | Conclusion

5.1. Summary of Findings

This research successfully developed an automated binary diagnostic system to distinguish between **Biotic** and **Abiotic** plant stresses. By implementing a customized **MobileNetV2** architecture paired with an **Adaptive Tile-based Voting Framework**, we effectively bypassed the "Resolution Bottleneck" common in traditional CNN pipelines.

The key findings are summarized as follows:

- **High Diagnostic Accuracy**: The optimal model state at **Epoch 12** achieved a weighted **F1-score of 0.95** and a leaf-level accuracy of **95.5%** on the multi-species test set.
- **Robust Feature Localization**: **Grad-CAM** visualizations confirmed that the model focuses on biologically relevant regions. High-attention zones (**Red/Orange**) strictly align with necrotic and chlorotic lesions, while background noise is successfully suppressed (**Blue/Purple**).
- **Generalization Potential**: The system demonstrated a robust cross-species inference accuracy of **83.97%**, validating its ability to learn universal pathological textures across diverse crop cultivars.

5.2. Future Work

Despite the high performance achieved, further research is required to transition this system from a laboratory setting to real-world agricultural deployment:

- **Dataset Expansion and Synthetic Augmentation**: To mitigate data scarcity in rare abiotic stress categories, future work will explore **Generative Adversarial Networks (GANs)** to synthesize realistic lesion patterns and increase dataset diversity.
- **Edge Device Optimization**: While MobileNetV2 is efficient, further quantization

and pruning are necessary to enable real-time, on-device inference on low-power mobile platforms used by farmers in the field.

- **Handling Co-occurrence and Early Detection**: Future iterations will focus on multi-label classification to identify **overlapping stresses** (e.g., simultaneous pest damage and nutrient deficiency) and improve sensitivity toward subtle, early-stage symptoms to enable preemptive intervention.

Bibliography

[1] Gao, Z., et al. (2020). Deep Learning Application in Plant Stress Imaging: A Review. *AgriEngineering*.

[2] Kaushik, P., et al. (2024). Enhanced Plant Stress Classification using DCNN with MobileNetV2. *ICACRS*.